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Chapter 1

Abstract

Food wastage is a common problem in college and institutional messes. Food is often prepared according to estimates, but actual attendance can vary because of absences, schedule changes, or other factors. This can result in excess food and unnecessary wastage.

The proposed **Mess Food Wastage Tracker and Management System** records meal attendance, food preparation, consumption, and wastage. It provides reports that help mess staff and administrators understand wastage patterns and make better decisions about future food preparation.

The system follows the **Agile Software Development Model**, allowing features to be developed in small increments and improved using feedback from students, mess staff, and administrators.

Chapter 2

Objective

The main objective is to reduce avoidable food wastage by maintaining accurate meal and food records.

The system aims to:

1. Record expected attendance for each meal.
2. Maintain daily menus and meal schedules.
3. Record prepared, consumed, and wasted food quantities.
4. Calculate wastage and wastage percentage.
5. Maintain historical records for comparison.
6. Generate daily, weekly, and monthly reports.
7. Help administrators make better food preparation decisions.

Chapter 3

Introduction

3.1 Purpose

This Software Requirements Specification defines the requirements and basic design of the Mess Food Wastage Tracker and Management System. It describes what the system should do, who will use it, and the main quality and operational requirements.

3.2 Scope

The system is intended for a college, hostel, or similar institutional mess. It will manage:

- student meal attendance,
- meal schedules and menus,
- food preparation records,
- food consumption records,
- food wastage records, and
- reports and historical data.

The system focuses on food wastage tracking and management. Full accounting, procurement, or hostel-management functions are outside the present scope.

3.3 Definitions

Term	**Meaning**
Student	User who may attend meals in the mess.
Mess Staff	Staff responsible for food preparation and recording meal data.
Administrator	User responsible for system and mess management.
Meal	A scheduled service such as breakfast, lunch, snacks, or dinner.
Expected Attendance	Number of students expected to attend a meal.
Prepared Quantity	Amount of food prepared for a meal.
Consumed Quantity	Amount of food consumed during a meal.
Wasted Quantity	Amount of prepared food that remains unused or is discarded.
Wastage Rate	Percentage of prepared food that is wasted.
Menu	Food items planned for a particular meal.
Report	Summary generated from stored system data.

3.4 System Overview

Students can view meal information and mark their expected attendance. Mess staff record preparation, consumption, and wastage data. Administrators manage users, meals, menus, and reports. The stored data is then used to calculate wastage and identify recurring patterns.

Chapter 4

SDLC Model

4.1 Selected Model: Agile

The **Agile Software Development Model** is selected for this project. Agile develops software through repeated iterations rather than delivering the entire system at once.

For this system, the first version can provide attendance and basic wastage recording. Features such as reports, analytics, notifications, and prediction can then be added and refined in later iterations.

4.2 Advantages

1. **Flexible:** Requirements can change based on user feedback.
2. **Incremental:** Features can be developed and tested in smaller parts.
3. **Early feedback:** Users can test features before the complete system is finished.
4. **Early delivery:** A basic working version can be available quickly.
5. **Easy improvement:** Problems can be identified and corrected during each iteration.

4.3 Disadvantages

1. Requires regular feedback from users.
2. Frequent changes can make planning difficult.
3. Poorly managed iterations may lead to insufficient documentation.
4. Continuous feature requests can cause scope creep.

5. Good communication between team members is necessary.

4.4 Why Agile Was Selected

The requirements of a mess-management system may become clearer only after users start using it. Students, mess staff, and administrators may have different expectations from the system.

Agile allows the team to first implement the essential features and then improve the system based on actual feedback. This makes it more suitable than a rigid development process for the proposed system.

A possible development sequence is:

1. Requirements and planning
2. User and meal management
3. Attendance management
4. Food and wastage recording
5. Reports and analytics
6. Testing and refinement

4.5 Agile Model Diagram

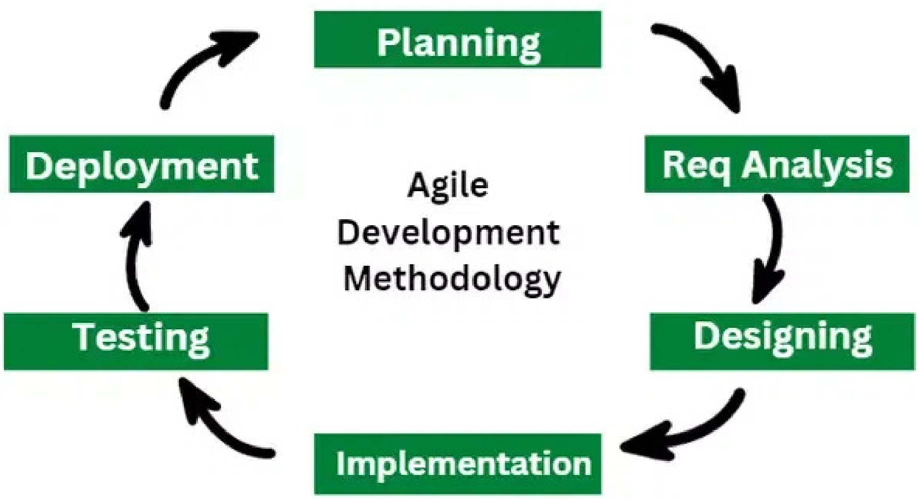

Figure 4.1: Agile Software Development Model

Chapter 5

Overall Description

5.1 Product Perspective

The proposed system is a standalone information-management system for an institutional mess. It receives information from students, mess staff, and administrators, stores it in a central database, and produces useful summaries and reports.

5.2 Main Functions

The main functions are:

1. User authentication and role management.
2. Meal and menu management.
3. Student meal attendance.
4. Food preparation and consumption recording.
5. Food wastage recording and calculation.
6. Historical analysis and report generation.

5.3 Users

5.3.1 Student

Students can view meals and menus, mark meal attendance, and modify their attendance within the allowed time.

5.3.2 Mess Staff

Mess staff record prepared, consumed, and wasted quantities and can view relevant meal information.

5.3.3 Administrator

Administrators manage users, meals, menus, configuration, and reports.

5.4 Operating Environment

The system may be implemented as a web application and accessed using a modern browser on a computer or mobile device. It may use a relational database and a local or cloud-based server.

5.5 Constraints

The main constraints are:

- The accuracy of reports depends on the accuracy of entered data.
- Students need access to the system to submit attendance.
- Network availability may be required.
- Mess staff may have limited time to enter data.
- User information must be protected from unauthorized access.

5.6 Assumptions

The system assumes that users provide reasonably accurate information, mess staff record food quantities regularly, users have valid accounts, and the mess follows a defined meal schedule.

Chapter 6

Requirement Specification

6.1 Functional Requirements

6.1.1 FR1 – Authentication and Authorization

1. The system shall allow users to log in using valid credentials.
2. The system shall identify the user's role after login.
3. The system shall restrict functions according to user role.
4. The administrator shall be able to activate or deactivate accounts.

6.1.2 FR2 – Meal Attendance

1. The system shall display available meals for a selected day.
2. A student shall be able to mark attendance for a meal.
3. A student shall be able to modify attendance before the deadline.
4. The system shall store the student, meal, and date for each attendance record.
5. The system shall calculate expected attendance for a meal.

6.1.3 FR3 – Menu Management

1. Authorized users shall be able to create and modify menus.
2. A menu shall be associated with a meal.
3. Previous menu information shall be retained as historical data.

6.1.4 FR4 – Food Recording

1. Mess staff shall be able to record prepared food quantity.
2. The system shall associate food records with a meal and food item.
3. Authorized users shall be able to correct incorrect records.

6.1.5 FR5 – Wastage Recording

1. Mess staff shall be able to record consumed and wasted quantities.
2. The system shall calculate the remaining and wasted quantity.
3. The system shall calculate the wastage percentage.
4. The system shall maintain historical wastage records.

6.1.6 FR6 – Analysis

1. The system shall calculate wastage for individual meals.
2. The system shall provide daily, weekly, and monthly summaries.
3. The system shall identify meals with high wastage.
4. The system shall support comparison of wastage across dates and meals.

6.1.7 FR7 – Reports

1. The system shall generate daily wastage reports.
2. The system shall generate weekly and monthly reports.
3. The system shall provide attendance and preparation summaries.

6.1.8 FR8 – Administration

1. The administrator shall manage student and staff accounts.
2. The administrator shall configure meal schedules.
3. The administrator shall manage food items and menus.
4. The administrator shall view system reports.

6.2 Non-Functional Requirements

6.2.1 NFR1 – Performance

1. Common operations should respond within a reasonable time.
2. Attendance submission should require minimal processing.
3. Reports should be generated within an acceptable time.

6.2.2 NFR2 – Security

1. Unauthorized users shall not access protected functions.
2. Passwords shall not be stored in plain text.
3. Users shall only access functions allowed by their roles.
4. Administrative operations shall require authorization.

6.2.3 NFR3 – Usability

1. The interface should be simple and understandable.
2. Common operations should require few steps.
3. Error and validation messages should be clear.

6.2.4 NFR4 – Reliability

1. Completed transactions should not result in data loss.
2. The system should handle invalid input safely.
3. Database backup should be supported.

6.2.5 NFR5 – Scalability

1. The system should support increasing numbers of users and records.
2. The architecture should allow additional mess facilities in the future.

6.2.6 NFR6 – Maintainability

1. The system should be divided into logical modules.
2. Major components should be documented.
3. Changes to one module should have limited impact on others.

6.2.7 NFR7 – Availability

The system should normally be available during mess operating hours. Planned maintenance should preferably be scheduled outside critical meal periods.

Chapter 7

Decision Analysis

7.1 Decision Tree

The decision tree determines whether preparation should be increased, maintained, or reduced using expected attendance and historical wastage.

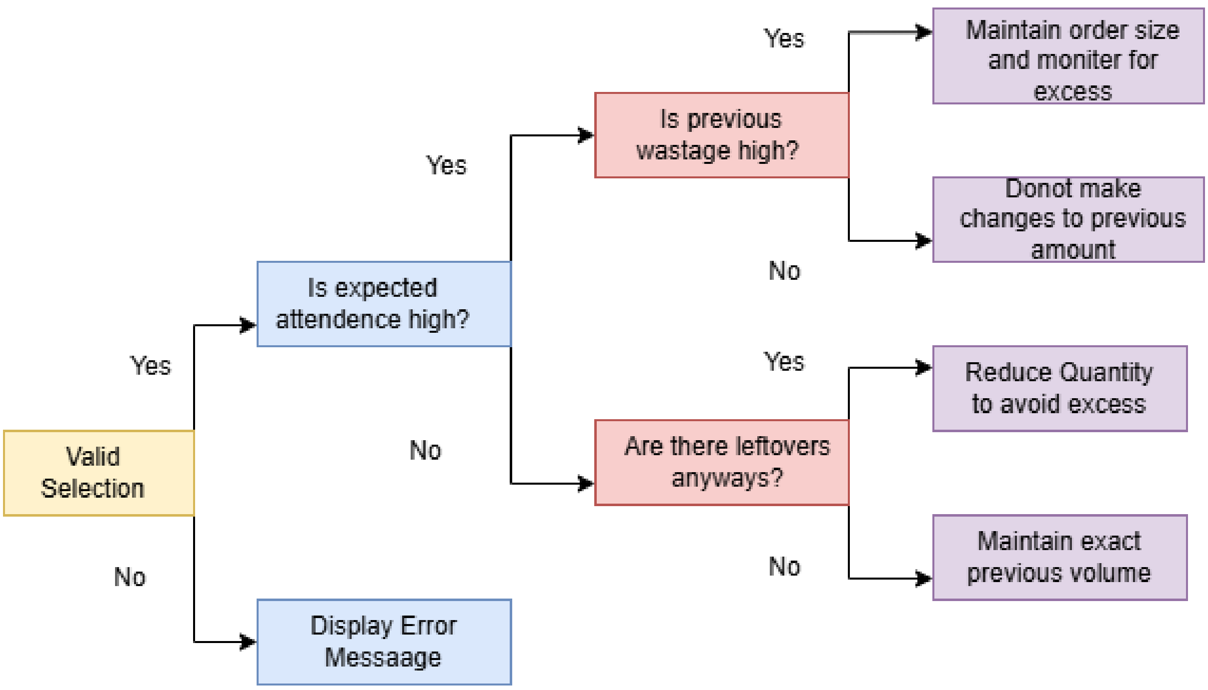

Figure 7.1: Decision Tree for Meal Preparation

7.2 Decision Table

The following table summarizes the same decision logic.

Table 7.1: Decision Table for Meal Preparation

Conditions / Actions	**R1**	**R2**	**R3**	**R4**	**R5**
Valid Selection?	No	Yes	Yes	Yes	Yes
Expected attendance high?	–	Yes	Yes	No	No
Previous wastage high?	–	Yes	No	–	–
Leftovers anyway?	–	–	–	Yes	No
Display error message	X				
Maintain order size and monitor excess		X			
Do not change previous amount			X		
Reduce quantity to avoid excess				X	
Maintain exact previous volume					X

Chapter 8

System Design

8.1 DFD Level 0

The Level 0 DFD treats the complete application as one process and shows its interaction with the three main external entities.

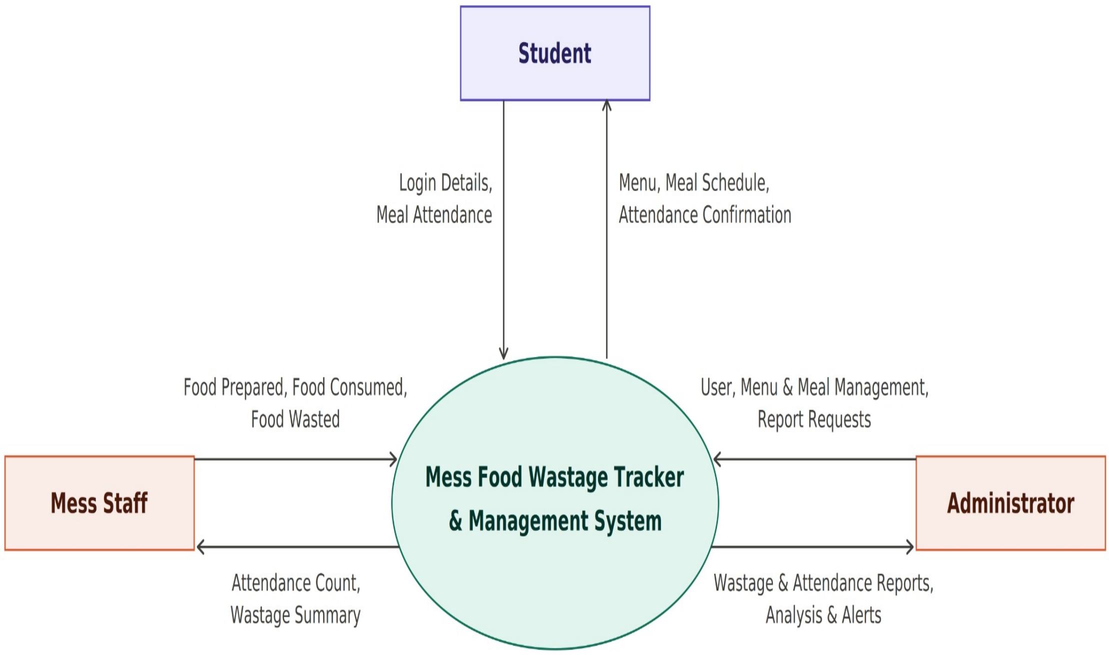

Figure 8.1: Data Flow Diagram Level 0

8.2 DFD Level 1

The Level 1 DFD decomposes the system into the following major processes:

1. User and Authentication Management
2. Meal and Menu Management
3. Attendance Management
4. Food Recording
5. Wastage Management
6. Report and Analytics Management

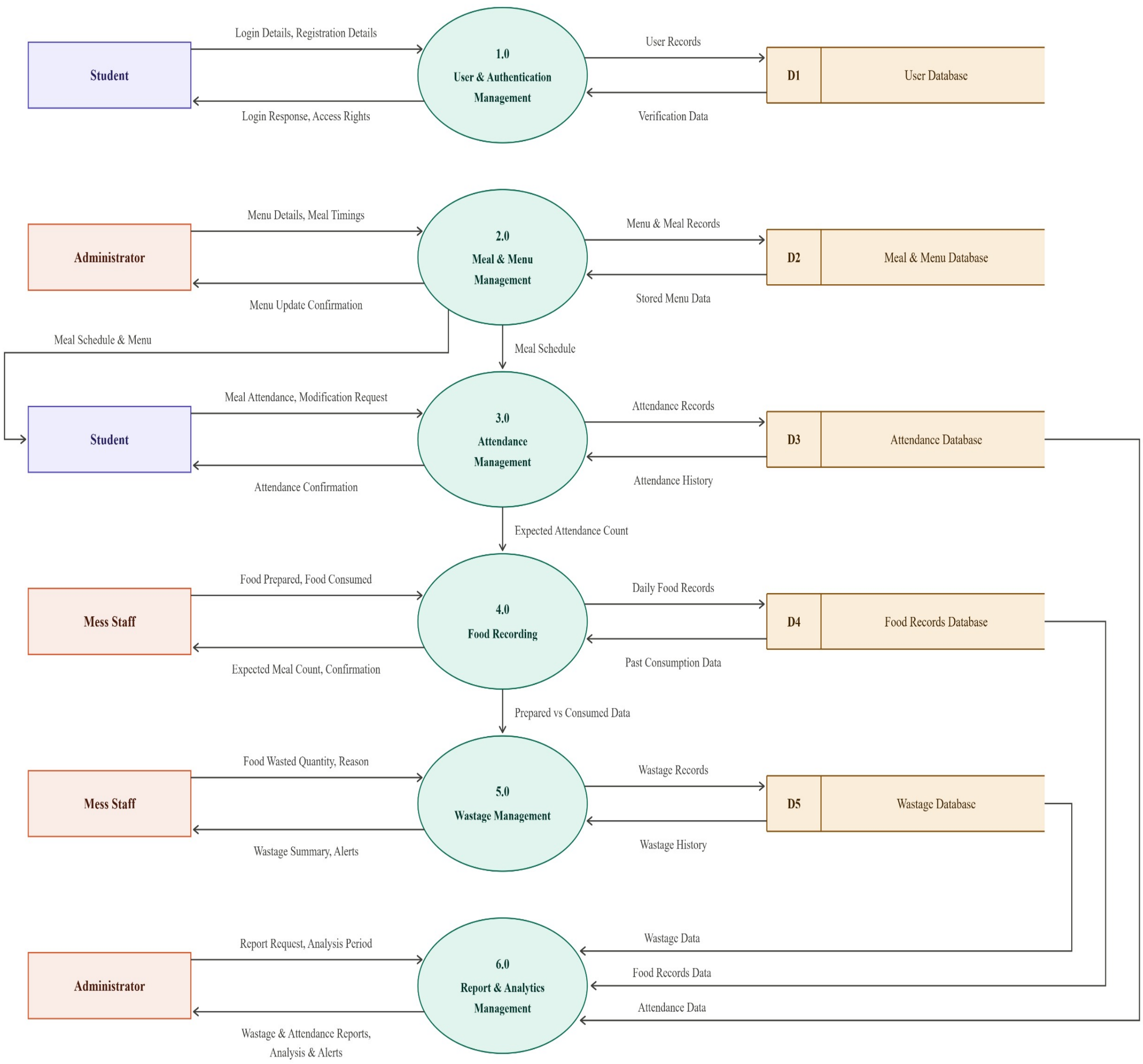

Figure 8.2: Data Flow Diagram Level 1

8.3 Structure Chart

The structure chart shows the hierarchy of the main software modules.

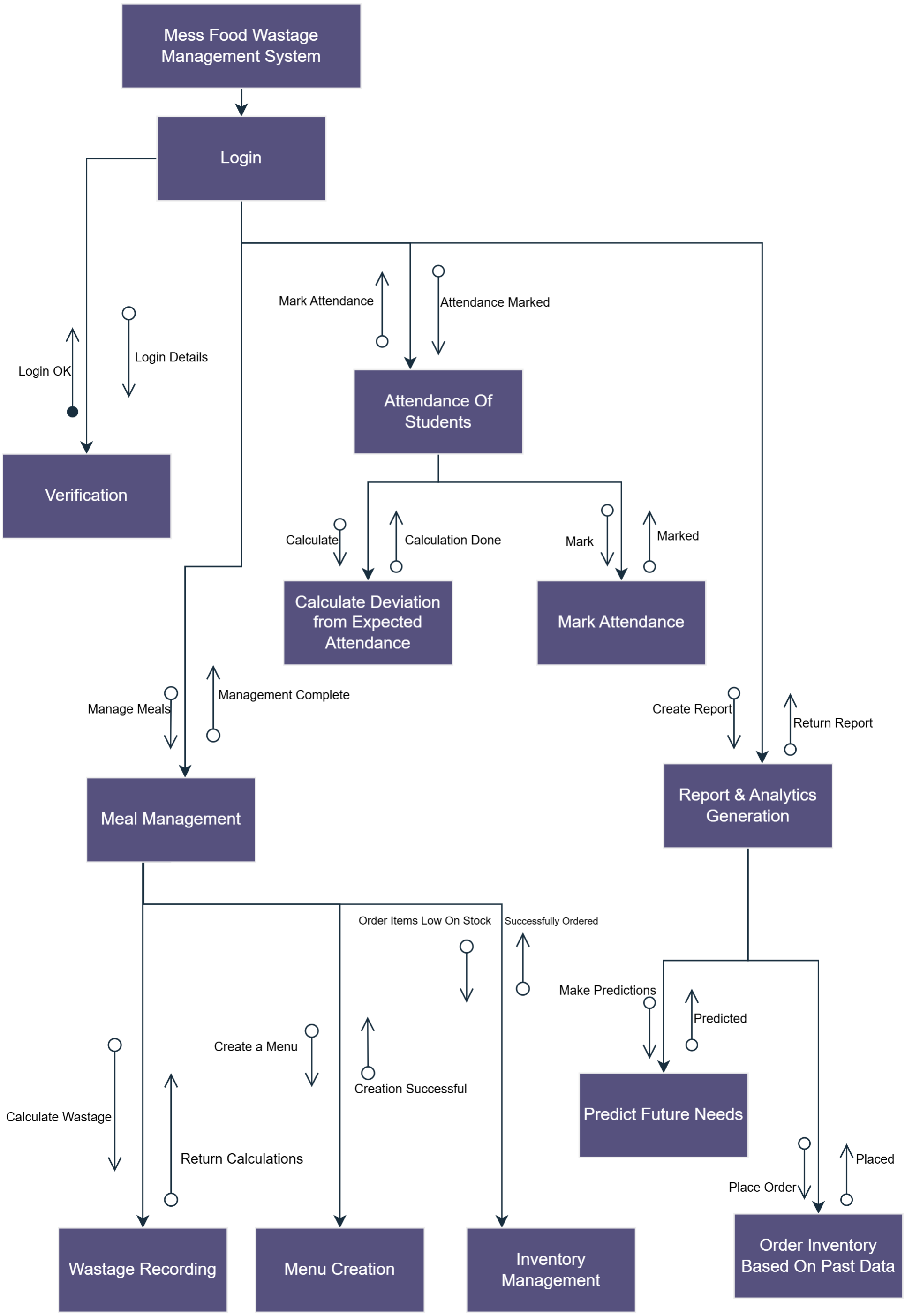

Figure 8.3: Structure Chart

8.4 Use Case Diagram

The main actors are Student, Mess Staff, and Administrator. The major use cases cover authentication, attendance, meal and menu management, food recording, wastage tracking, and reporting.

8.5 State Chart Diagram

The state chart represents the lifecycle of a meal from scheduling to report generation.

8.6 Class Diagram

The class diagram represents the main entities and their relationships.

The principal classes are *User*, *Student*, *MessStaff*, *Administrator*, *Meal*, *Menu*, *FoodItem*, *Attendance*, *FoodRecord*, *WastageRecord*, and *Report*.

Chapter 9

Behavioral Design

9.1 Activity Network Diagram

The activity network represents the main sequence of activities involved in handling a meal.

9.2 Interaction Diagram

The interaction diagram represents the communication between system objects when mess staff record food wastage.

The main objects are the Mess Staff, User Interface, Food Management, Wastage Management, and Database components.

9.3 Sequence Diagram

The sequence diagram shows the same wastage-recording operation in chronological order. It shows how messages move between the user interface, food management, wastage management, and database.

Chapter 10

Additional Requirements

10.1 Input Requirements

The system may receive:

- login credentials,
- meal attendance,
- meal and menu information,
- prepared quantity,
- consumed quantity,
- wasted quantity, and
- administrative settings.

10.2 Output Requirements

The system may produce:

- expected attendance,
- preparation records,
- wastage quantity and percentage,
- daily/weekly/monthly summaries,
- high-wastage meal information, and
- historical comparisons.

10.3 Data Requirements

The database should maintain information about users, roles, meals, menus, food items, attendance, preparation, consumption, wastage, and reports.

Chapter 11

Conclusion and Future Work

11.1 Conclusion

The Mess Food Wastage Tracker and Management System provides a structured way to record and monitor food wastage in an institutional mess. By combining attendance data with preparation and wastage records, the system can help staff understand where unnecessary waste occurs and support better meal planning.

Agile is suitable for the project because the system can be developed incrementally and improved through feedback from its users.

11.2 Future Work

Future versions may include:

1. Machine-learning based food-demand prediction.
2. A dedicated mobile application.
3. Automatic reminders for meal attendance.
4. Interactive dashboards and charts.
5. Cost estimation of wasted food.
6. Integration with inventory and procurement systems.
7. Support for multiple mess facilities.
8. Integration with weighing sensors for automatic measurement.

Chapter 12

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